

Quantum Computing and Quantum Machine Learning

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Plans for the week of April 6-10, 2026

- ▶ Discussions of the QPE algorithm and its implementations
- ▶ Material this week:
 - ▶ PDF slides at <https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week11/pdf/week11slides.pdf>
 - ▶ Jupyter-notebooks at <https://github.com/CompPhysics/QuantumComputingMachineLearning/tree/gh-pages/doc/pub/week11/ipynb>
- ▶ Project two suggestions, see <https://github.com/CompPhysics/QuantumComputingMachineLearning/tree/gh-pages/doc/Projects/2026/Project2>. Note that these suggestions will be revised till April 7
- ▶ Video of lecture at <https://youtu.be/Rsn2INejK7o>
- ▶ Whiteboard notes at <https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/HandWrittenNotes/2026/Lectureweek11.pdf>

What is the HHL algorithm

The HHL algorithm—named after Aram Harrow, Avinatan Hassidim, and Seth Lloyd—is one of the central quantum algorithms for solving linear systems of equations.

The core idea is to solve a linear system

$$Ax = b$$

What is the QAOA algorithm?

Next week, our plan is to discuss the QAOA algorithm. **QAOA** stands for the Quantum Approximate Optimization Algorithm.

Each part of the name reflects something essential about the method:

- ▶ Quantum: it runs on a quantum computer, using qubits and quantum gates.
- ▶ Approximate: it typically finds near-optimal solutions rather than exact ones.
- ▶ Optimization: it is designed to solve optimization problems (like MaxCut, scheduling, portfolio selection).
- ▶ Algorithm: it is a well-defined hybrid quantum–classical procedure.

QAOA is a variational quantum algorithm that uses a parameterized quantum circuit to search for good solutions to hard optimization problems.

Codes and plans for the coming two weeks (April 13-24)

This week (April 8), with respect to codes, we focus on the **implementation of the QPE algorithm and the QFT algorithm**. For the HHL algorithm our main focus, as discussed in the slides at <https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week11/pdf/week11slides.pdf>, will be on the theoretical aspects of the algorithm.

Next week we will discuss codes for the HHL algorithm applied to linear algebra problems and the solution of differential equations like the Poisson equation (week of April 13-17). We will also discuss the implementation and the theory behind the QAOA algorithm next week. The subsequent week we will link the HHL and QAOA algorithms to quantum machine learning and start with discussions quantum machine learning, with an emphasis on quantum support vector machines and quantum neural networks. These topics will be the main topics for the rest of the semester. Suggestions for project 2 will be presented April 8.

Plans for the rest of the semester, April

April 6-10, 2026.

- ▶ Finalizing the QPE discussions, with codes and formalism
- ▶ The HHL algorithm for solving linear algebra problems

April 13-17, 2026.

- ▶ Summing up the HHL discussions, codes and algorithms
- ▶ The QAOA algorithm

April 20-24, 2026 Quantum Machine Learning.

- ▶ Linking the HHL and the QAOA algorithm with quantum machine learning (QML)
- ▶ Basics of QML
- ▶ Quantum Support vector machines (QSVM) and classical SVMs

April 27-May 1, 2026 Quantum machine learning.

- ▶ QSVM, algorithms and codes
- ▶ Quantum neural networks (QNN)

Plans for the rest of the semester, May

May 4-8, 2026 Quantum Machine Learning.

- ▶ Quantum neural networks, codes and implementations

May 11-15, 2026.

- ▶ Quantum neural networks and quantum Boltzmann machines

May 18-22, 2026.

- ▶ Summary of course and discussion of and work on project 2